

CYBERSECURITY · FOUNDATION

Cybersecurity Foundations

Understand the threat. Build the defense.



Practical security foundations across identity, networks, applications and incident response.



8-12 weeks



4 modules



16 lessons

[Explore the course →](#)

Concepts, worked examples, assignments and a coherent course project.

WHO IT IS FOR · WHAT YOU WILL LEARN

Your next capability.

Develop security judgment through threat modeling, IAM, network and application security, cloud concepts, monitoring and incident response exercises.



Understand common attack surfaces



Apply secure development practices



Identify identity and network risks



Practice incident response thinking



Audience

- Developers
- QA engineers
- IT professionals
- Security beginners



Prerequisites

- Basic IT or software familiarity

FROM CONCEPT TO EVIDENCE



One curriculum, three connected views

This guide summarizes the same modules and lessons shown on the course overview and in the learning workspace. The next pages explain the scope and practical evidence for every module.

DETAILED MODULE CURRICULUM · 1-2

Build the foundations.

The module names, lesson sequence and evidence below match the course learning workspace.

1 Security Foundations

Review an isolated learning application using synthetic records and explicit authorization to test.

1.1 Security principles

Confidentiality, integrity and availability describe different security needs.

1.2 Threat actors and attack surfaces

An attack surface includes inputs, dependencies, interfaces and administrative access.

1.3 Threat modeling

A threat model connects actors, assets, trust boundaries and abuse cases.

1.4 Security controls

Preventive, detective and recovery controls serve different purposes.

You will produce: Threat model and authorized test plan.

2 Identity and Networks

Keep identity, permission and network exposure separate in the design.

2.1 Authentication and authorization

Authentication establishes who is calling; authorization decides what that caller may do.

2.2 IAM and least privilege

Least privilege limits permissions to the task and duration needed.

2.3 Network security

Network controls restrict reachability but do not replace application authorization.

2.4 Cloud security basics

Cloud security includes identity, configuration, secrets, data protection and operational monitoring.

You will produce: Access matrix and boundary test evidence.

Practice is part of every module

Work through the example, record your reasoning, complete the module assignment and compare your work with the review criteria.

DETAILED MODULE CURRICULUM · 3-4

Apply, integrate and demonstrate.

The module names, lesson sequence and evidence below match the course learning workspace.

3 Application Security

Validate application boundaries throughout development.

3.1 OWASP risks

Use risk categories as prompts for investigation, not a complete security checklist.

3.2 Web and API security

Validate requests and enforce object-level permission before returning private data.

3.3 Secrets and data protection

Keep secrets out of source code and browser bundles.

3.4 Secure development lifecycle

Build security review into design, implementation, testing and release.

You will produce: Application review and remediation report.

4 Monitoring and Response

Make security events useful for investigation and recovery.

4.1 Security logging

A log should identify the event, time, actor or service and decision without exposing unnecessary sensitive payloads.

4.2 Detection concepts

A detection rule maps observable behavior to a review signal.

4.3 Incident response

An incident process covers triage, containment, investigation, recovery and learning.

4.4 Security review capstone

The capstone presents scope, assets, findings, controls, residual risk and retest evidence.

You will produce: Security event plan, tabletop record and capstone report.

Practice is part of every module

Work through the example, record your reasoning, complete the module assignment and compare your work with the review criteria.

PROJECTS AND WORKING METHODS

Defensive security investigation

Work within an authorized lab, identify weaknesses, inspect evidence and develop a remediation and incident-response report.



Threat and risk assessment



Lab investigation evidence



Remediation and response playbook



Tools and methods

OWASP

Threat modeling

IAM

Security logging

Tools are learning choices, not partner endorsements.

Inside module 1: a worked example

Asset: private project notes

Boundary: browser -> API -> database

Abuse case: learner requests another learner's project

Control: server-side resource authorization

Evidence: authorized request succeeds; cross-user request is denied.

The exercise tests the boundary with synthetic accounts, not real users or third-party services.

How your work is reviewed

Show the artifact, explain your decisions, test the stated assumptions and identify limitations. Module assignments build toward the course project rather than replacing it with a single example.

DELIVERY, COMPLETION AND NEXT STEPS

Know what you are signing up for.

**Delivery**

Self-paced online · 8-12 weeks

Confirm current schedule, cohort arrangements and enrollment terms with the academy.

**Completion requirements**

Complete the required coursework and module assignments, and pass the course assessment. A course completion certificate is available after those requirements are met.

What the learning workspace includes**Follow the modules**

Expand the curriculum and open the exact lesson you need.

**Apply the concepts**

Read explanations, inspect examples and complete module practice.

**Keep your evidence**

Use resources, notes, discussion and assignments to support your learning.

Learning commitments, not outcome guarantees

Completion is not a placement, employment or performance guarantee. The course project is educational evidence; any production use requires appropriate review and validation.

Start with the curriculum. Continue with the first lesson.

Review your starting point, download this guide and explore the connected course experience.

[Go to the first lesson](#) →

Client-review edition. The static learning workspace saves demonstration progress locally; production enrollment, assessment, communications and certificate issuance are not connected.